

690 7050 1092

สูตรนี้ 15 นาที = 1 ชั่วโมง

- a) 1024×1024 array of 32 bit number = $1024 \times 1024 \times 4 \text{ bytes} = 2^{22} \text{ bytes}$
- memory = 2^{22} bytes
- $\frac{32}{8} = 4 \text{ bytes}$

9th main memory ବାଇଟ୍ = 1 M bytes = 2^{20} bytes

Page faults = $\frac{2^{30}}{2^{12}} = 2^8$ pages # ans

b) $10 \times 2^8 = 10240 \text{ ms}$ #ans

1)

Kunahya Khuntipong
63070501010

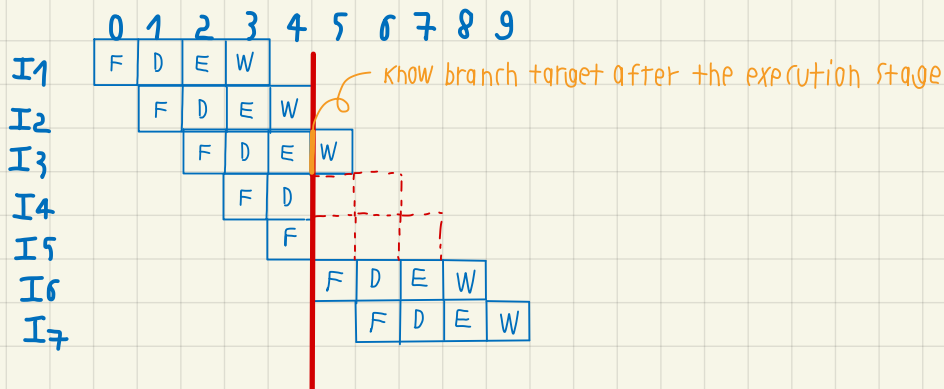
Question #1 (20 Min)

Consider the following code:

	Load #0, C	// I1
	Compare A,B	// I2 (A-B)
	Branch>0 Action1	// I3
	DEC C	// I4
	Branch NEXT	// I5
Action1	INC C	// I6
NEXT	Load #0, D	// I7

- a) Assume that a branch-not-taken is predicted, and that the branch target is known after the execution stage, draw a 4-stage pipeline (F-D-E-W) diagram when $A = 2$ and $B = 1$.
- b) Does delay slot exist? If so, show how to eliminate it?

a) The 4-stage pipeline is



- b) The delay slot exists.
To eliminate it, reorder the sequence of instruction.

	LOAD #0, C	
	COMPARE A, B	// (A-B)
	BRANCH > 0 ACTION1	
	LOAD #0, D	
	DEC C	
	BRANCH NEXT	
ACTION1	INC C	
NEXT	...	

2)

Kunahya Khunthipong
63070501010

Question #2 (30 Min)

"A 1024×1024 array of 32-bit numbers is stored in the main memory starting at address 0. The array is to be *normalized* as follows. For each column, the largest value is found and all elements of the column are divided by this maximum value. Assume that each page in the **direct-mapped cache** consists of 4K bytes, and the cache size is 1M bytes."

- How many page faults would occur if the elements of the array are stored in column order in the main memory?
- Suppose it takes 40 ms to load a page from the main memory to cache and time to load data from cache is very low, estimate the total time needed to perform this *normalization*.

100% Not sure Ka T-T
So hard jung lopy Aj.
(can I get Attemp point?!!)

a) Each row : 1024

so the number of page fault is 1024 #

b) We know that it takes 40 ms to load a page
so the estimated total time needed to
perform this normalization is $40\text{ms} \times 1024$
 $= 40.96 \text{ s}$ #

3)

Kunahya Khuntiptong
63070501010

Question #3 (40 Min)

This computer system uses set-associative (2 sets) mapping for cache memory (with LRU). Two figures show the source code and the cache content after running the code (capture after pass: $j=3,7,9$ and $i=4,2,0$).

```
SUM := 0
for j:=0 to 9 do
    SUM := SUM + A(0,j);
end
AVE := SUM / 10;
for i:=9 downto 0 do
    A(0,i) := A(0,i) / AVE;
end
```

Contents of data cache after pass :

Block position	j=3	j=7	j=9	i=4	i=2	i=0
Set 0 { 0	A(0,0)	A(0,4)	A(0,8)	A(0,4)	A(0,4)	A(0,0)
1	A(0,1)	A(0,5)	A(0,9)	A(0,5)	A(0,5)	A(0,1)
2	A(0,2)	A(0,6)	A(0,6)	A(0,6)	A(0,2)	A(0,2)
3	A(0,3)	A(0,7)	A(0,7)	A(0,7)	A(0,3)	A(0,3)
Set 1 { 4						
5						
6						
7						

Change one line of code to >> for i:=0 to 9 do

What's the content of cache should be? (draw the new table with the idea of above table)

The new table is

Contents of data cache

Block position	j=3	j=7	j=9	i=4	i=2	i=0
Set 0 { 0	A(0,0)	A(0,4)	A(0,8)	A(0,2)	A(0,6)	A(0,6)
1	A(0,1)	A(0,5)	A(0,9)	A(0,3)	A(0,7)	A(0,7)
2	A(0,2)	A(0,6)	A(0,6)	A(0,0)	A(0,4)	A(0,8)
3	A(0,3)	A(0,7)	A(0,7)	A(0,1)	A(0,5)	A(0,9)
Set 1 { 4						
5						
6						
7						

With LRU

(This table show the content after pass $j=3,7,9$ and $i=3,7,9$)

Online Quiz #2: Lecture 7-11: 3 points with 10 minutes per question

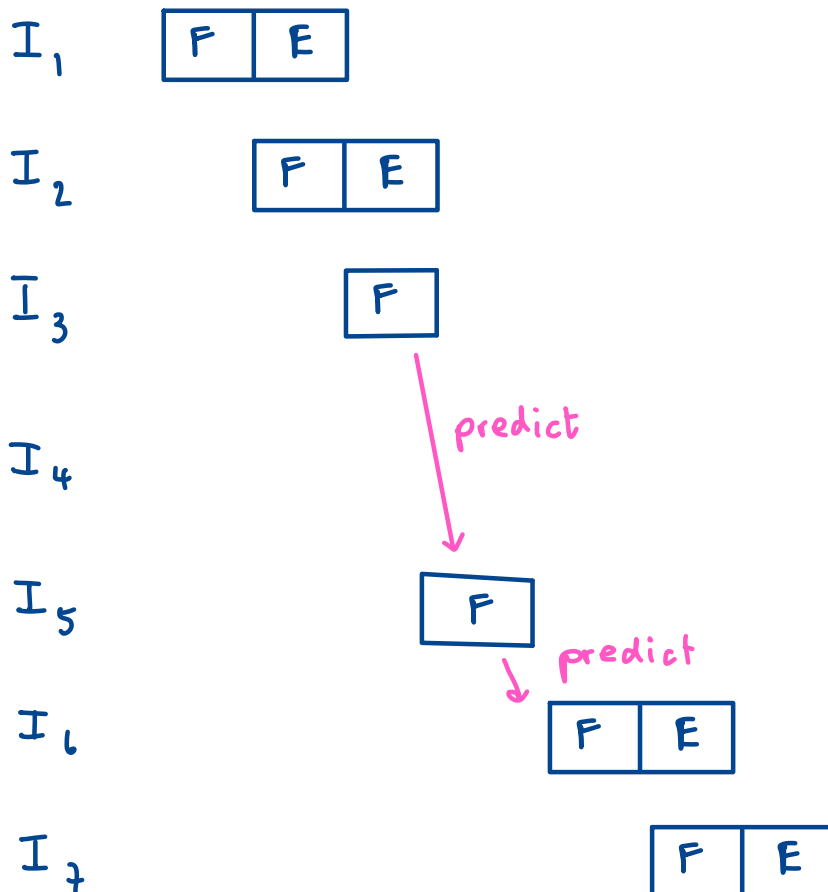
1) Consider the following code:

```

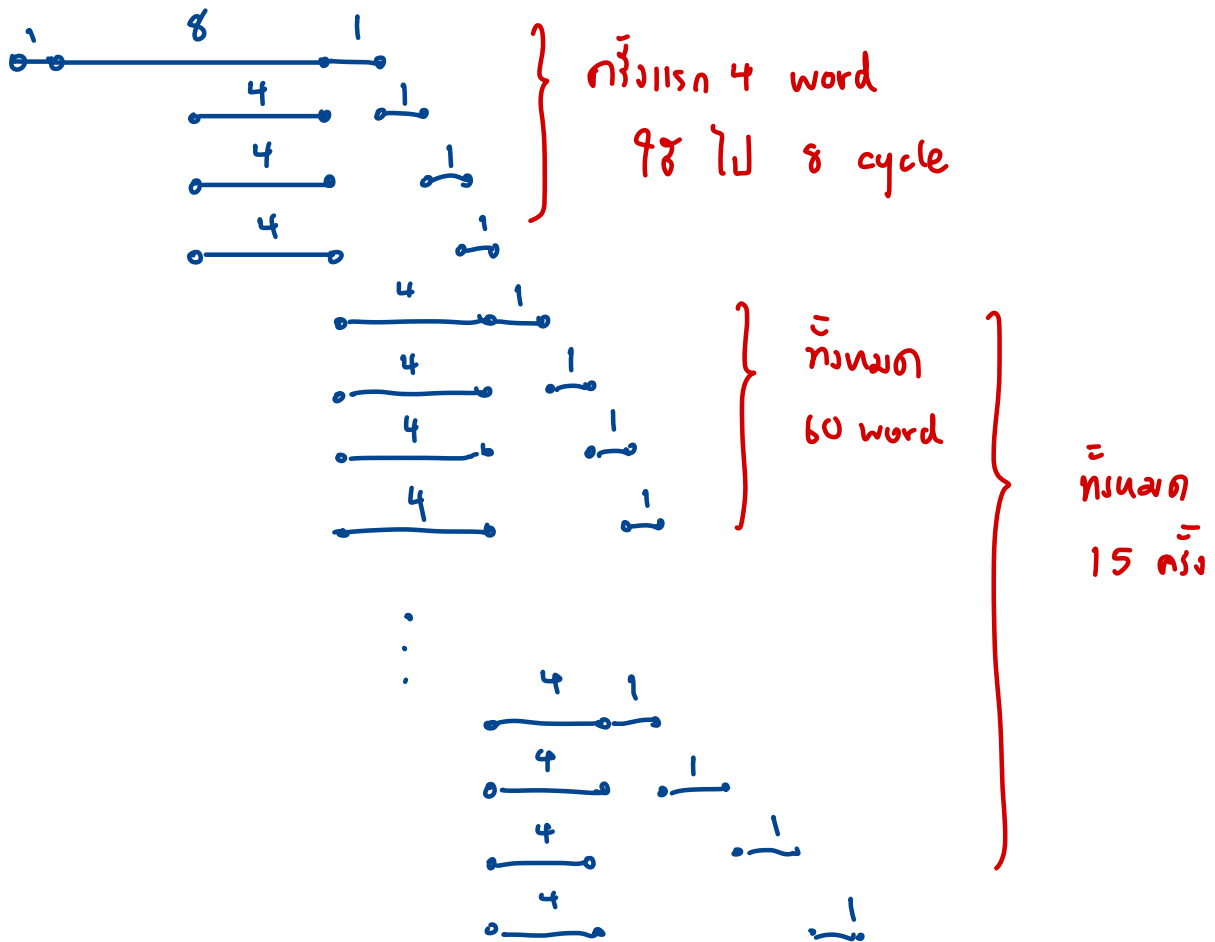
Load #0, C           // I1
Compare A,B          // I2 (A-B)
Branch>0 Action1     // I3
DEC C                // I4
Branch NEXT          // I5
Action1 INC C         // I6
NEXT Load #0, D       // I7

```

Assume that a **branch-not-taken** is predicted, and that the branch target is known after the execution stage, **draw** a 2-stage pipeline (Fetch-Execute) diagram when $A = 2$ and $B = 1$.



- 2) A memory system has the following properties: Hardware takes 1 clock cycle to send an address to the main memory. Memory latency is 8 cycles, but the subsequent word can be accessed in 4 cycles/word. One clock cycle is needed to send one word to the cache. Suppose the memory is constructed as *four interleaved modules*, what is the total time needed to load a block of 64 words from the memory? (Show the solution)



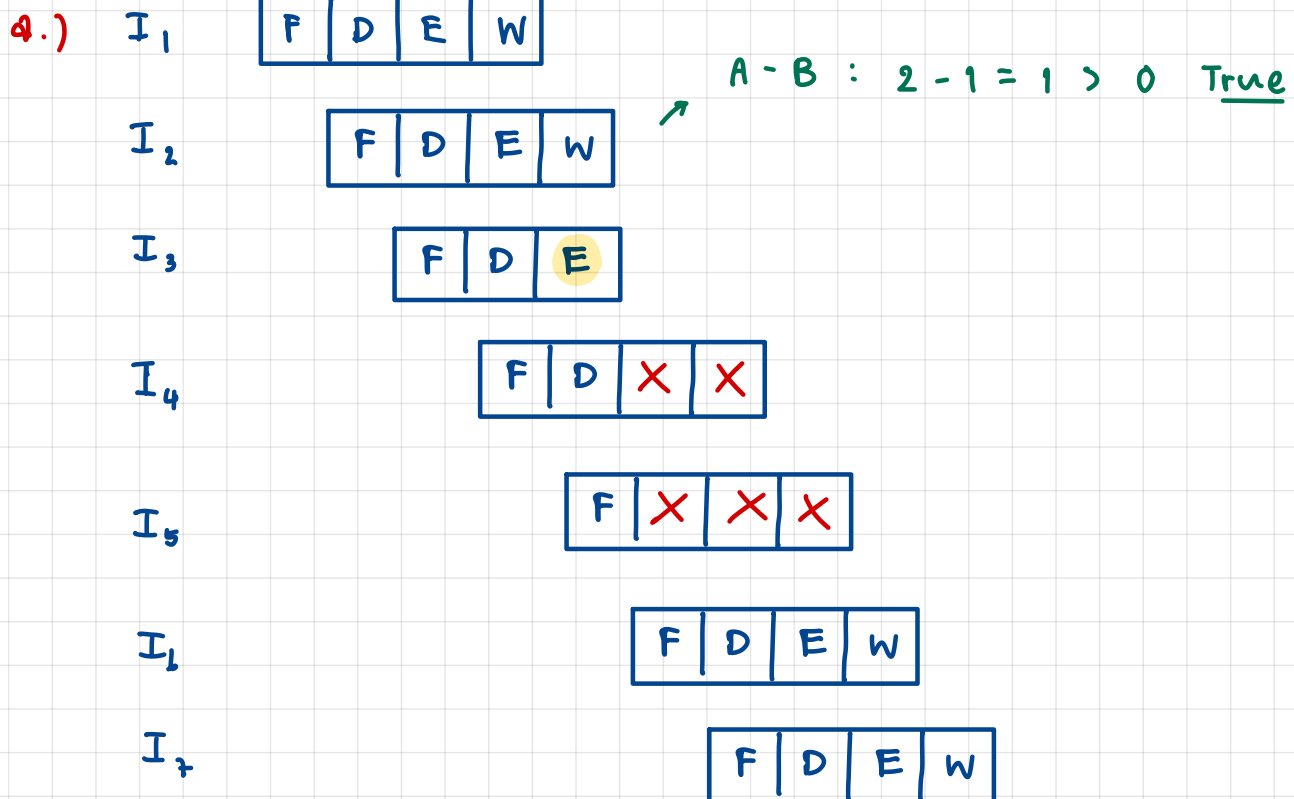
$$1 + 8 + 15 \times 8 = 129 \text{ cycles} \quad \#$$

Question #1 (20 Min)

Consider the following code:

	Load #0, C	// I1	
	Compare A,B	// I2 (A-B)	$2 - 1 = 1 > 0$
	Branch>0 Action1	// I3	
	DEC C	// I4	
	Branch NEXT	// I5	
Action1	INC C	// I6	
NEXT	Load #0, D	// I7	

- a) Assume that a branch-not-taken is predicted, and that the branch target is known after the execution stage, draw a 4-stage pipeline (F-D-E-W) diagram when $A = 2$ and $B = 1$.
- b) Does delay slot exist? If so, show how to eliminate it?



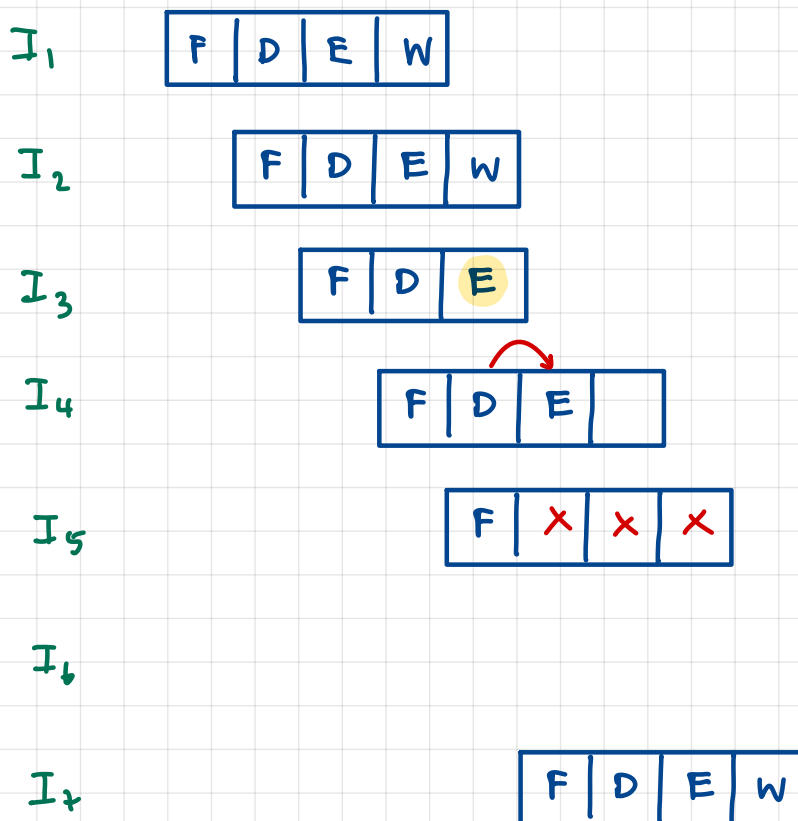
b.) จากข้อ a. : มี penalty จากการที่มี 2 inst. ถูกตัดทิ้ง

	Load #0, C	// I1
	Compare A,B	// I2 (A-B)
	Branch>0 Action1	// I3
	DEC C	// I4
	Branch NEXT	// I5
Action1	INC C	// I6
NEXT	Load #0, D	// I7

↪ ขั้ว

แก้ไข :

	Load #0, C	I ₁
	Compare A, B	I ₂
	Branch > 0 Action 1	I ₃
	Load #0, D	I ₄
	DEC C	I ₅
	Branch NEXT	I ₆
Action 1	INC C	I ₇
NEXT		I ₈

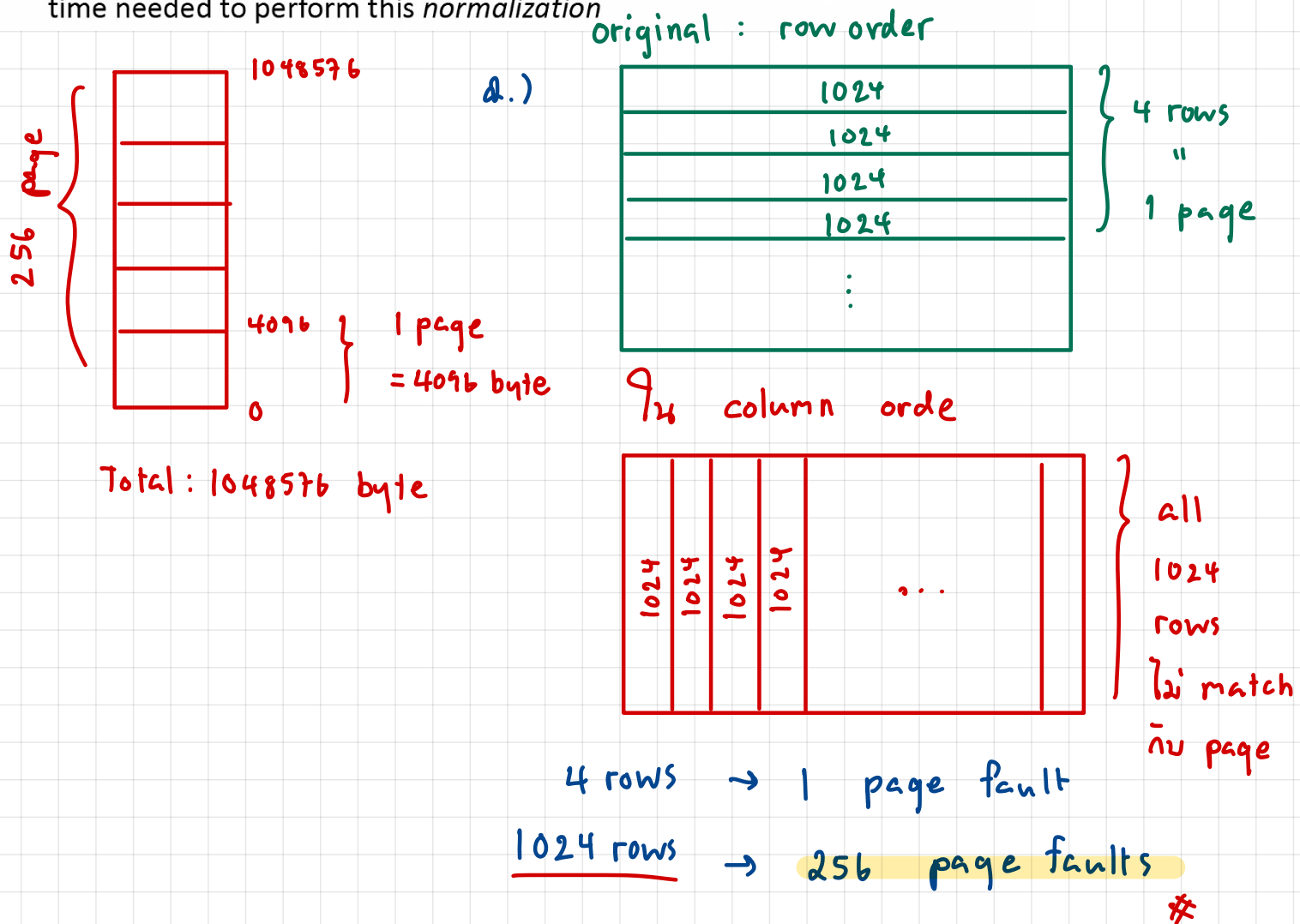


Question #2 (30 Min)

4 bytes

"A 1024x1024 array of 32-bit numbers is stored in the main memory starting at address 0. The array is to be *normalized* as follows. For each column, the largest value is found and all elements of the column are divided by this maximum value. Assume that each page in the **direct-mapped cache** consists of 4K bytes, and the cache size is 1M bytes."

- a) How many page faults would occur if the elements of the array are stored in column order in the main memory?
- b) Suppose it takes 40 ms to load a page from the main memory to cache and time to load data from cache is very low, estimate the total time needed to perform this *normalization*



b.) ① 4 row → 1 page

1024 row → 256 page *

② 1 page 98 40 ms

256 page 98 256 × 40 = 10240 ms approx. 10.24 วินาที *

Question #3 (40 Min)

This computer system uses set-associative (2 sets) mapping for cache memory (with LRU). Two figures show the source code and the cache content after running the code (capture after pass: $j=3,7,9$ and $i=4,2,0$).

```
SUM := 0
for j:=0 to 9 do
    SUM := SUM + A(0,j);
end
AVE := SUM / 10;
* for i:=9 downto 0 do
    A(0,i) := A(0,i) / AVE;
end
```

Contents of data cache after pass :

Block position	j = 3	j = 7	j = 9	i = 4	i = 2	i = 0
Set 0	A(0,0)	A(0,4)	A(0,8)	A(0,4)	A(0,4)	A(0,0)
1	A(0,1)	A(0,5)	A(0,9)	A(0,5)	A(0,5)	A(0,1)
2	A(0,2)	A(0,6)	A(0,6)	A(0,6)	A(0,2)	A(0,2)
3	A(0,3)	A(0,7)	A(0,7)	A(0,7)	A(0,3)	A(0,3)
Set 1						
4						
5						
6						
7						

Change one line of code to $\gg$ for $i:=0$ to 9 do

What's the content of cache should be? (draw the new table with the idea of above table)

	j	0	1	2	3	4	5	6	7	8	9
cache											
Set 1	0	A(0,0)	A(0,0)	A(0,0)	A(0,0)	A(0,0)	A(0,0)	A(0,0)	A(0,0)	A(0,8)	A(0,8)
	1			A(0,2)	A(0,2)	A(0,2)	A(0,2)	A(0,2)	A(0,2)	A(0,2)	A(0,2)
	2					A(0,4)	A(0,4)	A(0,4)	A(0,4)	A(0,4)	A(0,4)
	3							A(0,6)	A(0,6)	A(0,6)	A(0,6)
Set 2	4		A(0,1)	A(0,1)	A(0,1)	A(0,1)	A(0,1)	A(0,1)	A(0,1)	A(0,1)	A(0,1)
	5				A(0,3)	A(0,3)	A(0,3)	A(0,3)	A(0,3)	A(0,3)	A(0,3)
	6						A(0,5)	A(0,5)	A(0,5)	A(0,5)	A(0,5)
	7								A(0,7)	A(0,7)	A(0,7)

cache \ i		0	1	2	3	4	5	6	7	8	9
set 1	0	$A(0,8)$	$A(0,8)$	$A(0,8)$	$A(0,8)$	$A(0,8)$	$A(0,8)$	$A(0,6)$	$A(0,6)$	$A(0,6)$	$A(0,6)$
	1	$A(0,0)$	$A(0,0)$	$A(0,0)$	$A(0,0)$	$A(0,0)$	$A(0,0)$	$A(0,0)$	$A(0,0)$	$A(0,8)$	$A(0,8)$
	2	$A(0,4)$	$A(0,4)$	$A(0,2)$	$A(0,2)$	$A(0,2)$	$A(0,2)$	$A(0,2)$	$A(0,2)$	$A(0,2)$	$A(0,2)$
	3	$A(0,6)$	$A(0,6)$	$A(0,6)$	$A(0,6)$	$A(0,4)$	$A(0,4)$	$A(0,4)$	$A(0,4)$	$A(0,4)$	$A(0,4)$
set 2	4	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,7)$	$A(0,7)$	$A(0,7)$
	5	$A(0,3)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$	$A(0,1)$
	6	$A(0,5)$	$A(0,5)$	$A(0,5)$	$A(0,3)$	$A(0,3)$	$A(0,3)$	$A(0,3)$	$A(0,3)$	$A(0,3)$	$A(0,3)$
	7	$A(0,7)$	$A(0,7)$	$A(0,7)$	$A(0,7)$	$A(0,7)$	$A(0,5)$	$A(0,5)$	$A(0,5)$	$A(0,5)$	$A(0,5)$